

International Finance

Lecture VI – March 24th

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

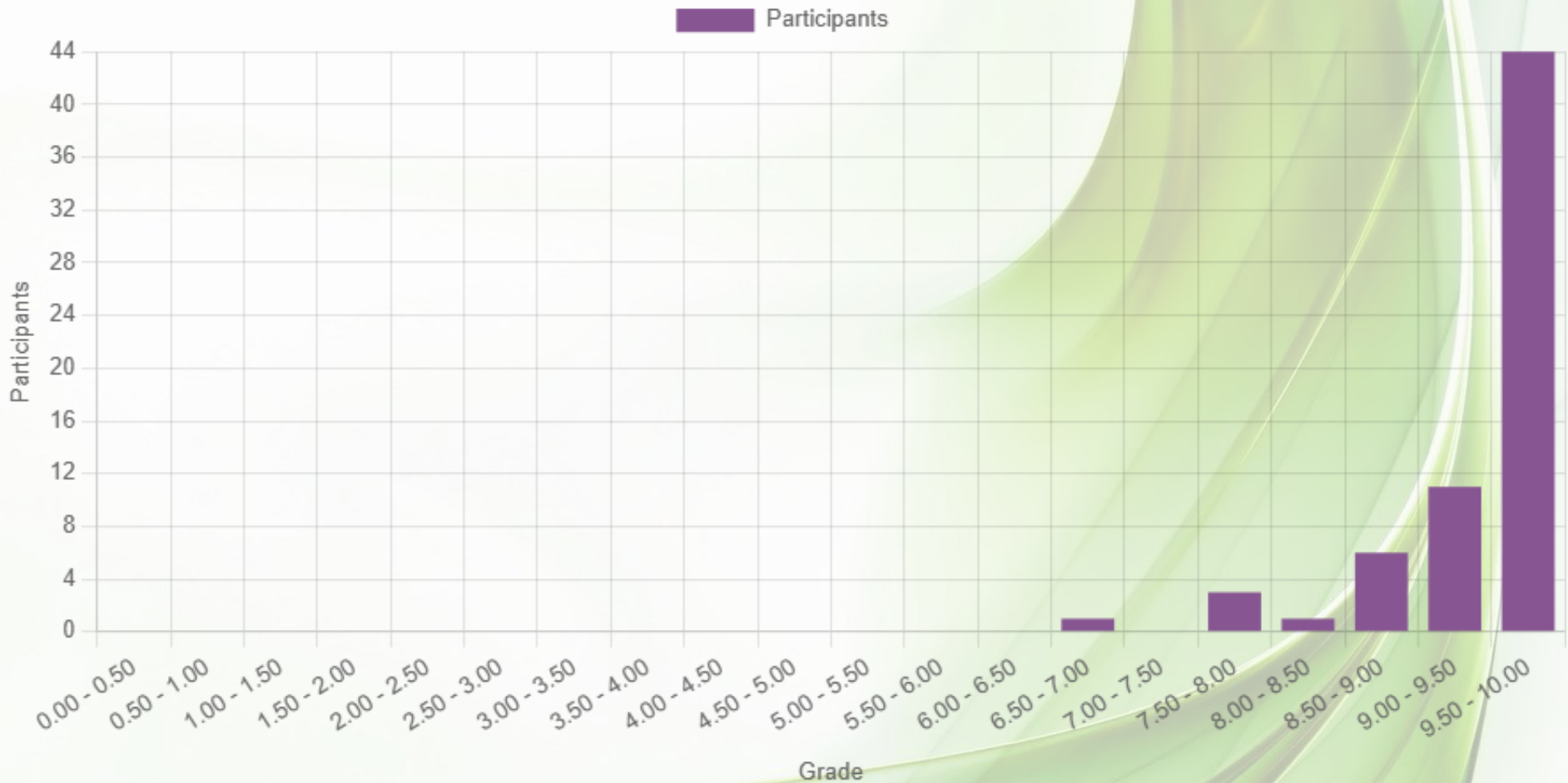
Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. (February 24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. Introduction to exchange rate determination: asset approach. **Covered** and uncovered **interest rate parity**. (March 10th)
5. **Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity.** Overshooting. (March 17th)
6. **Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect.** Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

Outline

- Course organization
 - Attendance, quiz, assignments (Teams + topics)
 - Quiz #6: will be open by tomorrow (Wednesday) midnight
 - The usual deadline: next Monday midnight
- Futures markets....
- Finishing the PPP and UIP issues: (International) Fisher effect
- Nominal and real effective exchange rates
- Explaining the deviations from the PPP and observed trends in exchange rates
 - Reminder: empirical evidence
 - Balassa-Samuelson model
 - Other explanations: Bhagwati
 - Empirical issues:
 - CEE countries and real appreciation
 - Real convergence in the EU

Quiz #5



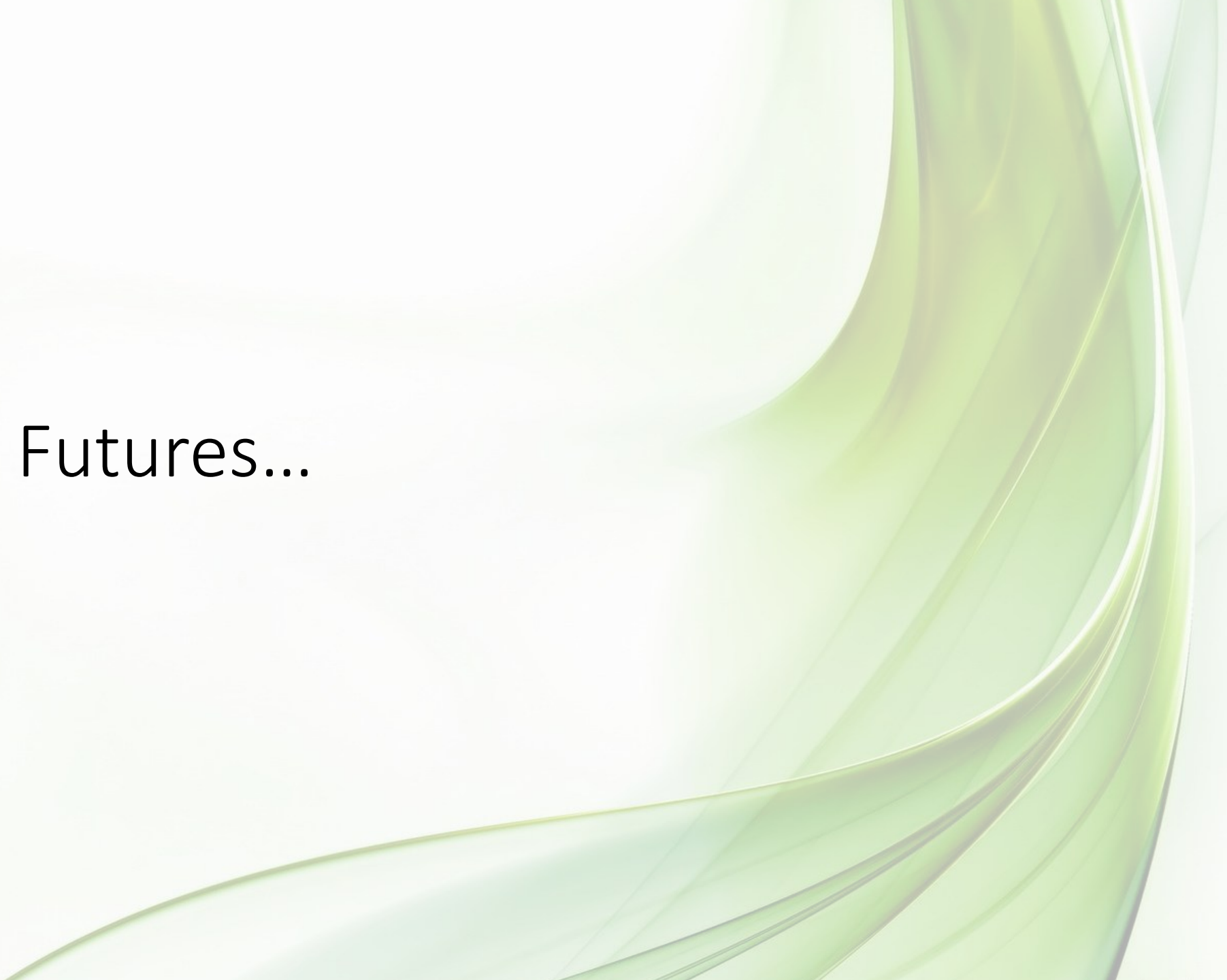
Average: 9.34

72 attempts/71 participants

Reading for the Next Week:

- Maurer (2023): The Divergence of Price Levels in the European Union
- Krugman & Obstfeld & Melitz: International Economics (chapter 17)
- Worldbank blog: Behind the numbers: How we measure global poverty
 - <https://blogs.worldbank.org/en/opendata/behind-the-numbers--how-we-measure-global-poverty>
- If you would like to play with data on price levels and price differences, you can try the WB ICP:
 - <https://databank.worldbank.org/source/icp-2021/>.

Futures...





Some “Footnotes”

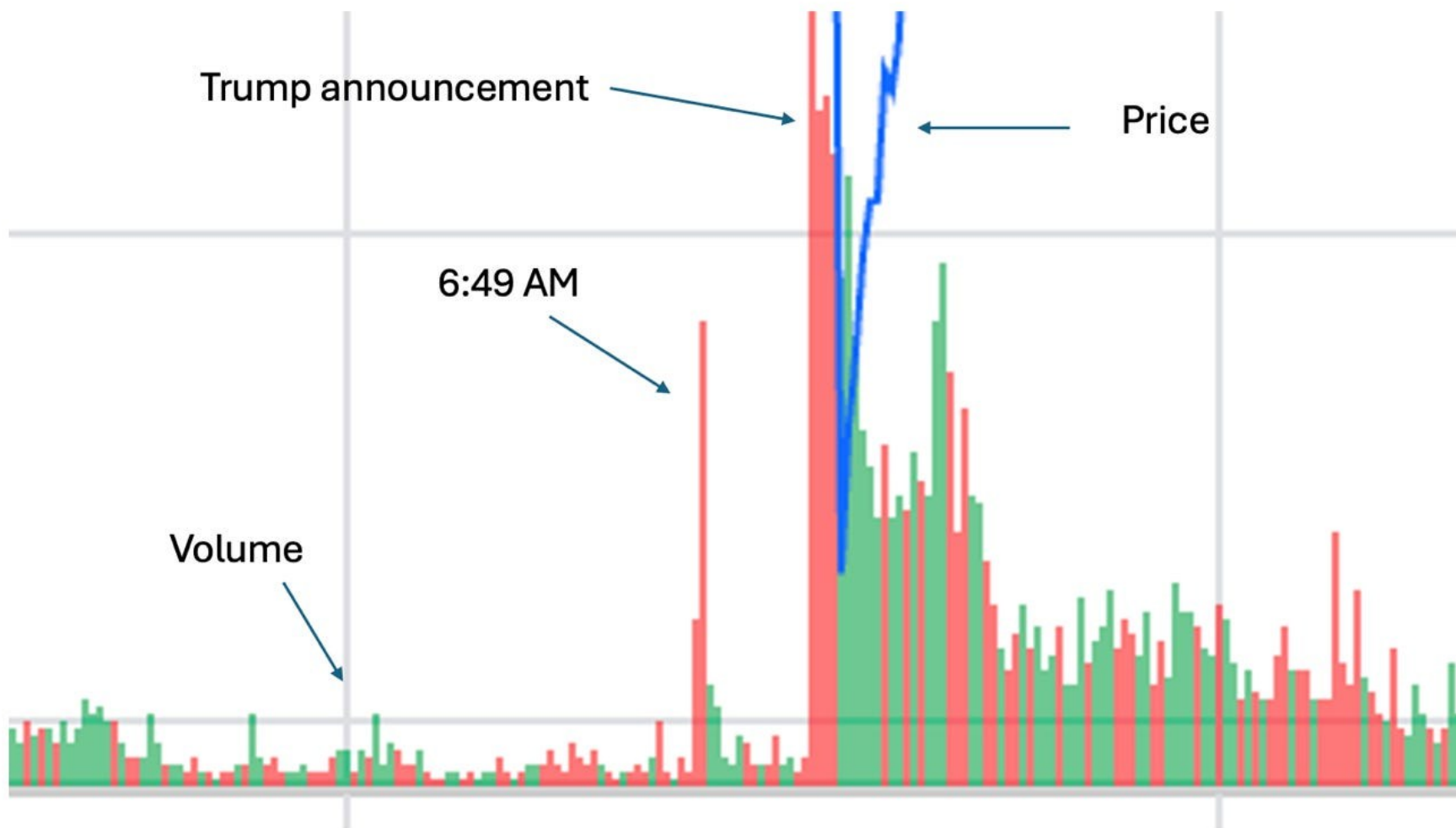
- WTI CFD, or West Texas Intermediate Contract for Difference is a financial derivative that allows traders to speculate on the price movements of WTI Crude Oil without actually owning the physical commodity. CFDs are popular in financial markets because they allow investors to benefit from both up and down market conditions.
- It is important to understand that CFD itself is not an asset but a derivative of different financial assets. In the financial markets depending on the provider, you can trade CFDs for instruments like stocks, indexes, ETFs, bonds, commodities, precious metals, cryptocurrencies, and forex.
- Investors can enter into a WTI CFD by agreeing to exchange the difference in the price of WTI crude oil between the opening and closing of the contract. This type of trading instrument enables participants to leverage their positions, potentially amplifying both gains and losses.
- For example, if a trader believes that the price of WTI crude oil will rise, they can enter a long WTI CFD position. Conversely, if they anticipate a decline in prices, they can go short. The profit or loss is then calculated based on the difference between the entry and exit prices.

Interesting Events in Futures Markets....

- Background:
 - Donald Trump declared an ultimatum to Iran – they had to open the Strait of Hormuz within 48 hours (deadline: Monday evening, Washington)
 - **At 7:05 a.m. Trump called the ultimatum off.**
- According to CNBC (and P. Krugman):
 - At around **6:50 a.m.** in New York, S&P 500 e-Mini futures trading on the CME recorded a sharp and isolated jump in volume, breaking from an otherwise subdued premarket backdrop. With thin liquidity typical of early trading hours, **the sudden burst stood out as one of the largest volume moments of the session up to that point.**
 - A similar pattern was observed in oil markets. **West Texas Intermediate May futures also saw a noticeable pickup in trading activity at roughly the same time, with a distinct volume spike interrupting otherwise quiet conditions.**

From: <https://paulkrugman.substack.com/p/treason-in-the-futures-markets>

Oil futures trading, morning of Mar. 23, 2026



*Back to our parities, how about
combining some of them?*

International Fisher Effect

(International) Fisher Effect (1)

- How about linking **the relative PPP** and interest rates (UIP)?
- Condition of interest rate parity (UIP) between dollar and euro asset:

$$R_{\$} = R_{\epsilon} + \left(E_{\$/\epsilon}^e - E_{\$/\epsilon} \right) / E_{\$/\epsilon} \quad (1)$$

- This should hold in both the long and short run
- **Long run:** relative PPP should hold too:

$$\frac{\left(E_{\$/\epsilon,t} - E_{\$/\epsilon,t-1} \right)}{E_{\$/\epsilon,t-1}} = \pi_{US,t} - \pi_{EMU,t}$$

(International) Fisher Effect (2)

- If the original relative PPP holds, it should also hold for expected inflation rates:

$$\frac{(E_{\$/\epsilon}^e - E_{\$/\epsilon})}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_{EMU}^e \quad (2)$$

- Combining (1) and (2) (and rearranging) leads to:

$$R_{\$} - R_{\epsilon} = \pi_{US}^e - \pi_{EMU}^e$$

- Long run relationship between ongoing inflation and interest rates:
 - **Ceteris paribus, a rise in a country's expected inflation rate will eventually cause an equal rise in the interest rate that deposits of its currency offer.**
 - Similarly, a fall in the expected inflation rate will eventually cause a fall in the interest rate

(International) Fisher Effect (3)

- Implications of Fisher effect:
 - If U.S. inflation were to rise permanently from a constant level of 5 % per year to a constant level of 10 % per year, dollar interest rates would eventually catch up with the higher inflation (and increase by 5 percentage points).
 - The real rate of returns on dollar assets (measure in U.S. goods and services) would be unchanged!
- Interesting aspect:
 - A currency depreciates if its interest rates rise relative to foreign interest rates
 - Why?
 - Long run equilibrium: a rise in the difference between home and foreign interest rates occurs only when expected home inflation rises (relative to the foreign interest rates).

Nominal and Real Exchange Rates

“Types” of Exchange Rates (2)

- Nominal v. **real exchange rates**

- Nominal rates: relative prices of currencies
- **Real exchange rates**: indicators based on data on nominal exchange rates and inflation rates of the respective countries
 - **Relative price of commodities/services!**
 - Used as a simple indicator of **changes in price competitiveness**

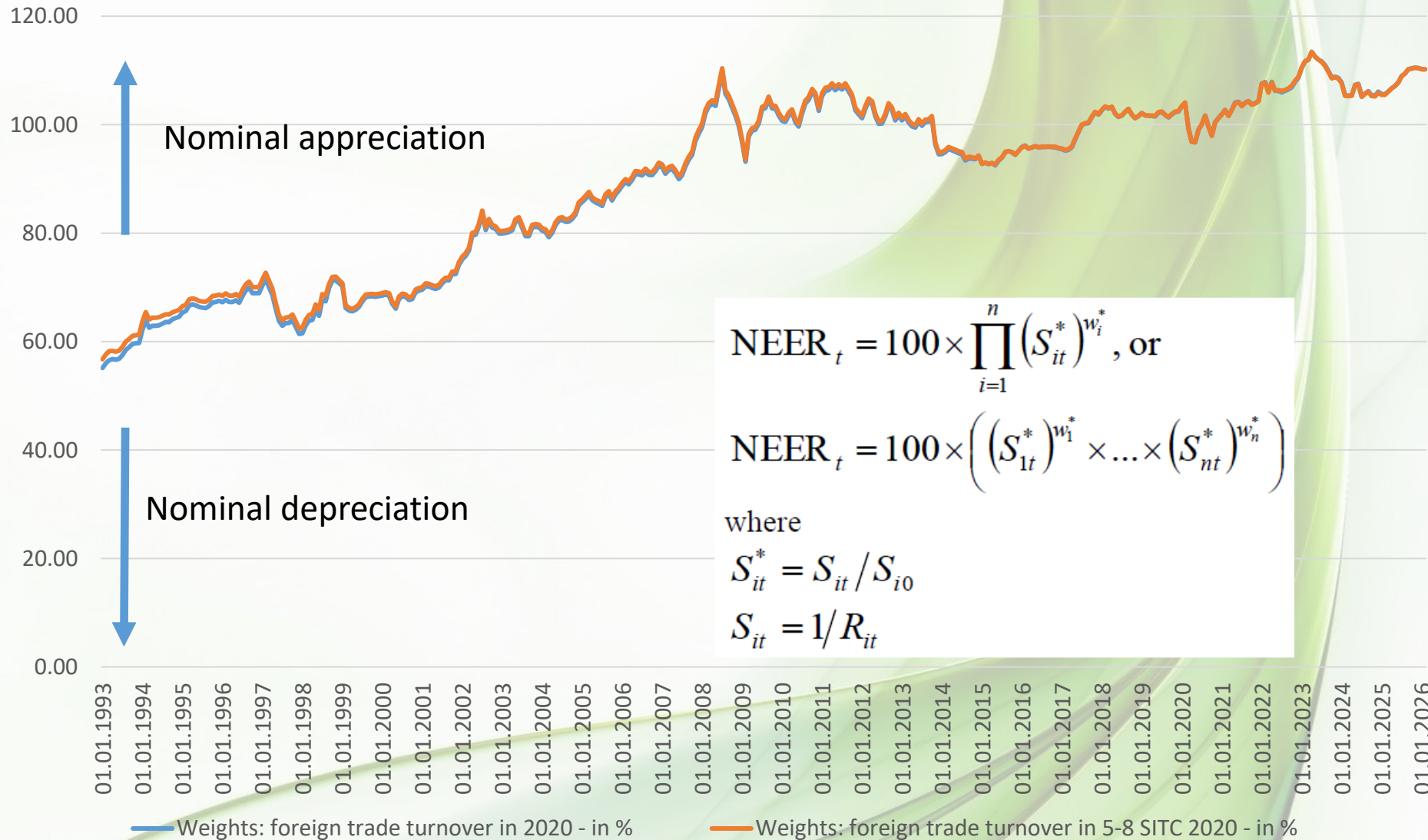
$$R = S \frac{P^*}{P} = \frac{SP^*}{P}$$

* Indicates foreign variables

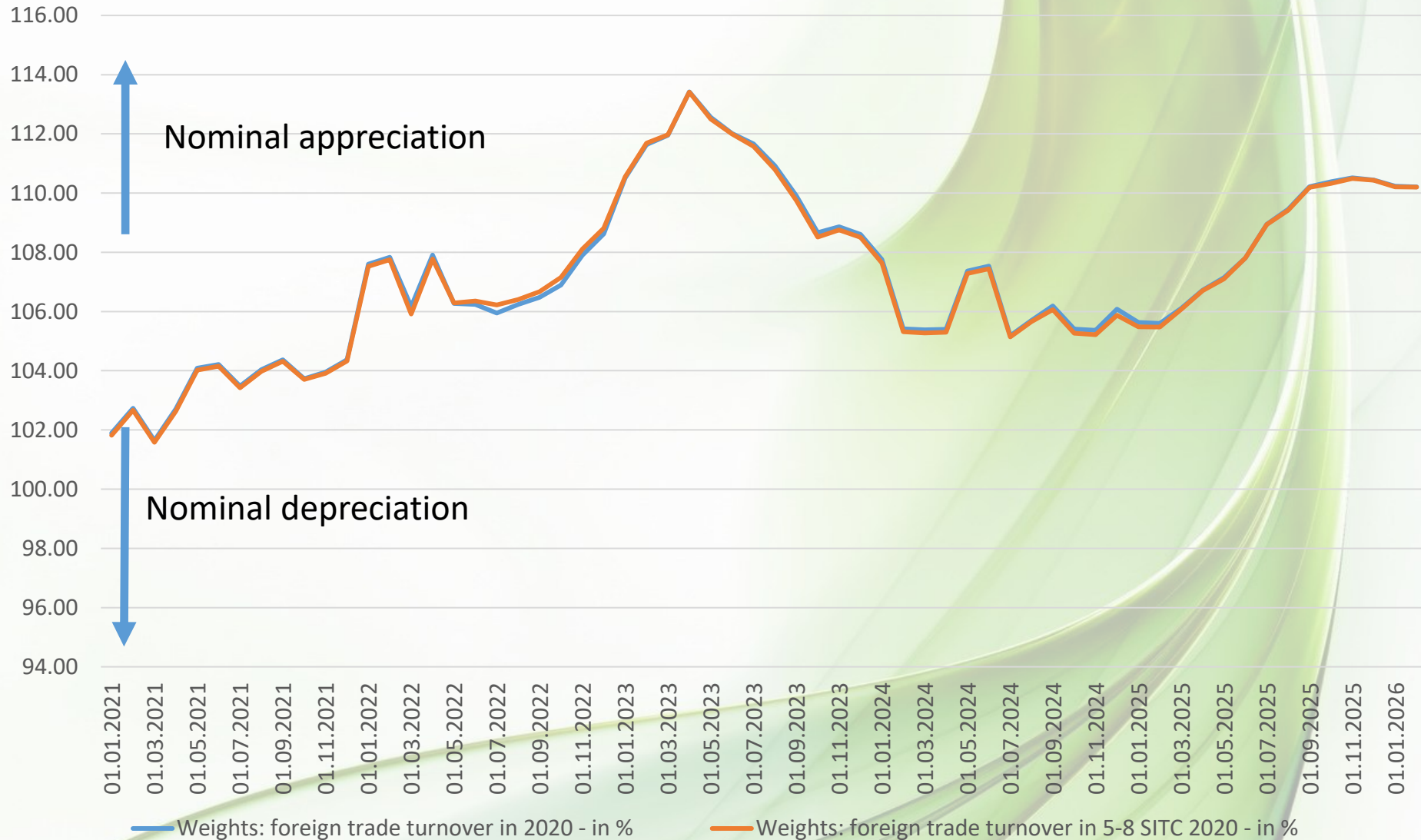
Many Versions of Real Effective Exchange Rates Indices

- What can differ?
 - Number of currencies considered
 - Weights
 - Price indices
- Example: REER by CNB
 - Quite a few options:
 - 2 sets of weights (total foreign trade or SITC 5-8)
 - 5 types of price indices:
 - Price indices (PPI and CPI, monthly)
 - Unit labor costs (whole economy x manufacturing only, quarterly)
 - GDP deflator (quarterly)

Example: Indices of Nominal Effective Exchange Rates by CNB (Base 2020)



Example: Indices of Nominal Effective Exchange Rates by CNB (Base 2020)



Source: CNB, Arad

REER: Formula Used by the CNB

$$REER_t = 100 \times \prod_{i=1}^n \left(\frac{S_{it}^*}{P_{it}^*} \right)^{w_i^*}$$

- where:

- S_{it}^* - basic index of the domestic currency to the currency of the i-th trading partner in period t
- P_{it}^* - ratio of the basic price index of the i-th trading partner in period t to the basic price index of the Czech Republic in period t, the base year being the same as for the calculation of S_i^*
- w_i^* - standardised weights of the currency of the i-th trading partner

REER by CNB: Current Weights

Based on 2020 weights for 13 currency areas

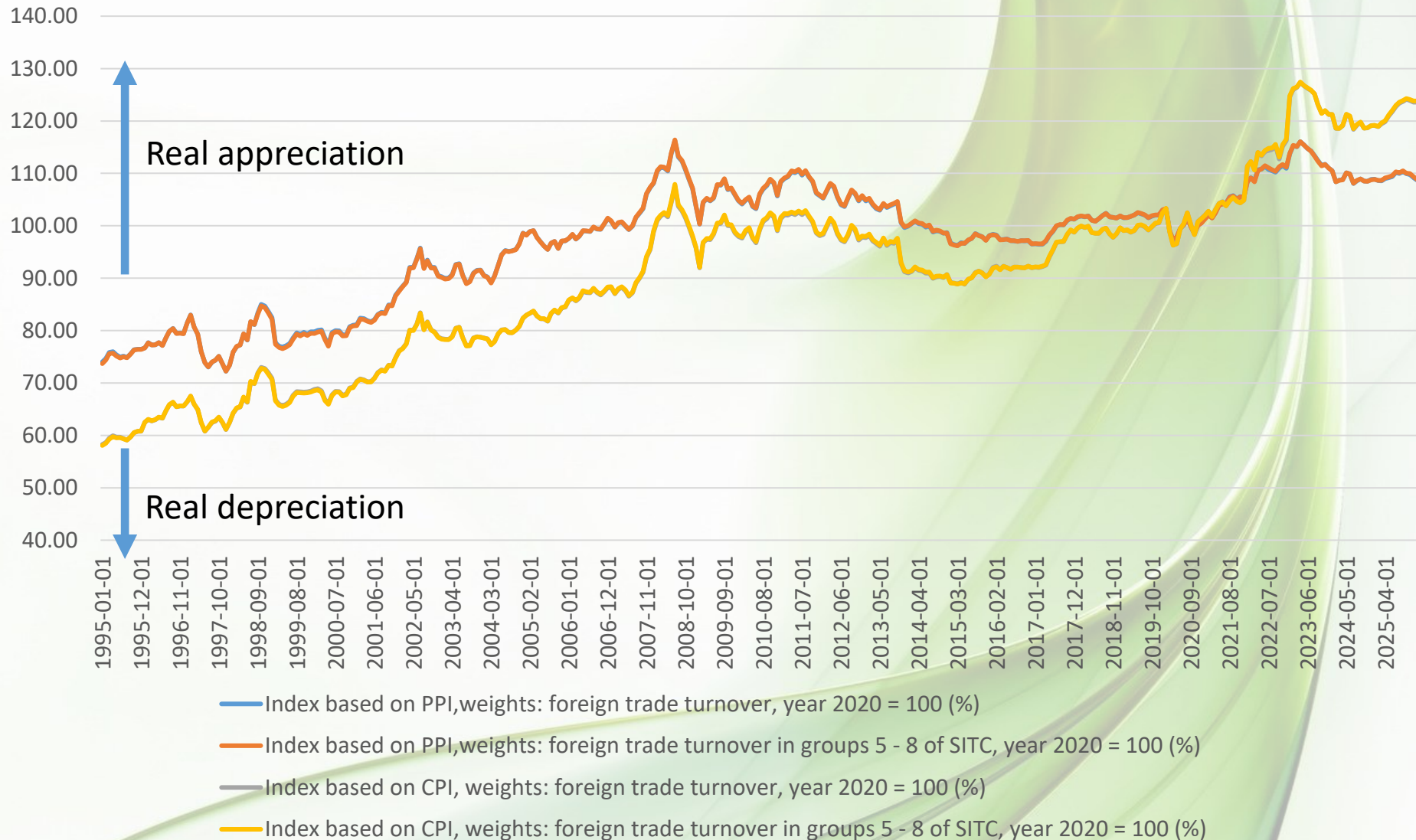
Weights of monetary areas calculated by share of total trade turnover of the Czech Republic

Variant I.	in %
Monetary area	2020
1 eurozone	64,0
2 Poland	8,5
3 China	7,5
4 Hungary	3,5
5 United Kingdom	3,2
6 USA	2,8
7 Russian Federation	2,5
8 Romania	1,8
9 Switzerland	1,4
10 Korea	1,4
11 Sweden	1,4
12 Japan	1,2
13 Denmark	0,8
Total	100,0

Weights of monetary areas calculated by share of total trade turnover of the Czech Republic for SITC groups 5-8

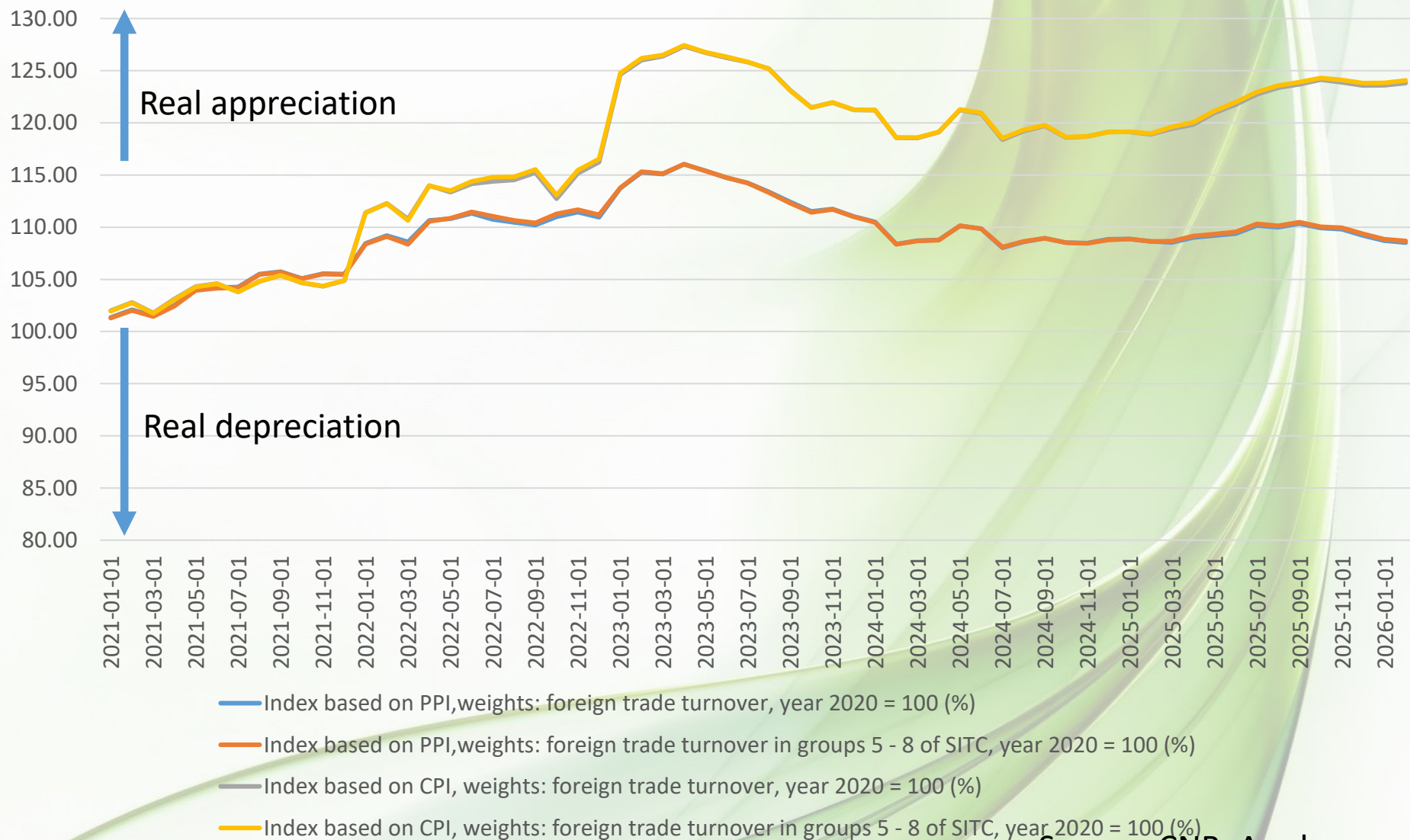
Variant II.	in %
Monetary area	2020
1 eurozone	64,0
2 Poland	8,4
3 China	7,8
4 Hungary	3,6
5 United Kingdom	3,2
6 USA	2,8
7 Russian Federation	1,9
8 Romania	1,8
9 Switzerland	1,5
10 Korea	1,5
11 Sweden	1,5
12 Japan	1,2
13 Denmark	0,8
Total	100,0

Example: Indices of Real Effective Exchange Rates by CNB (Base 2020)

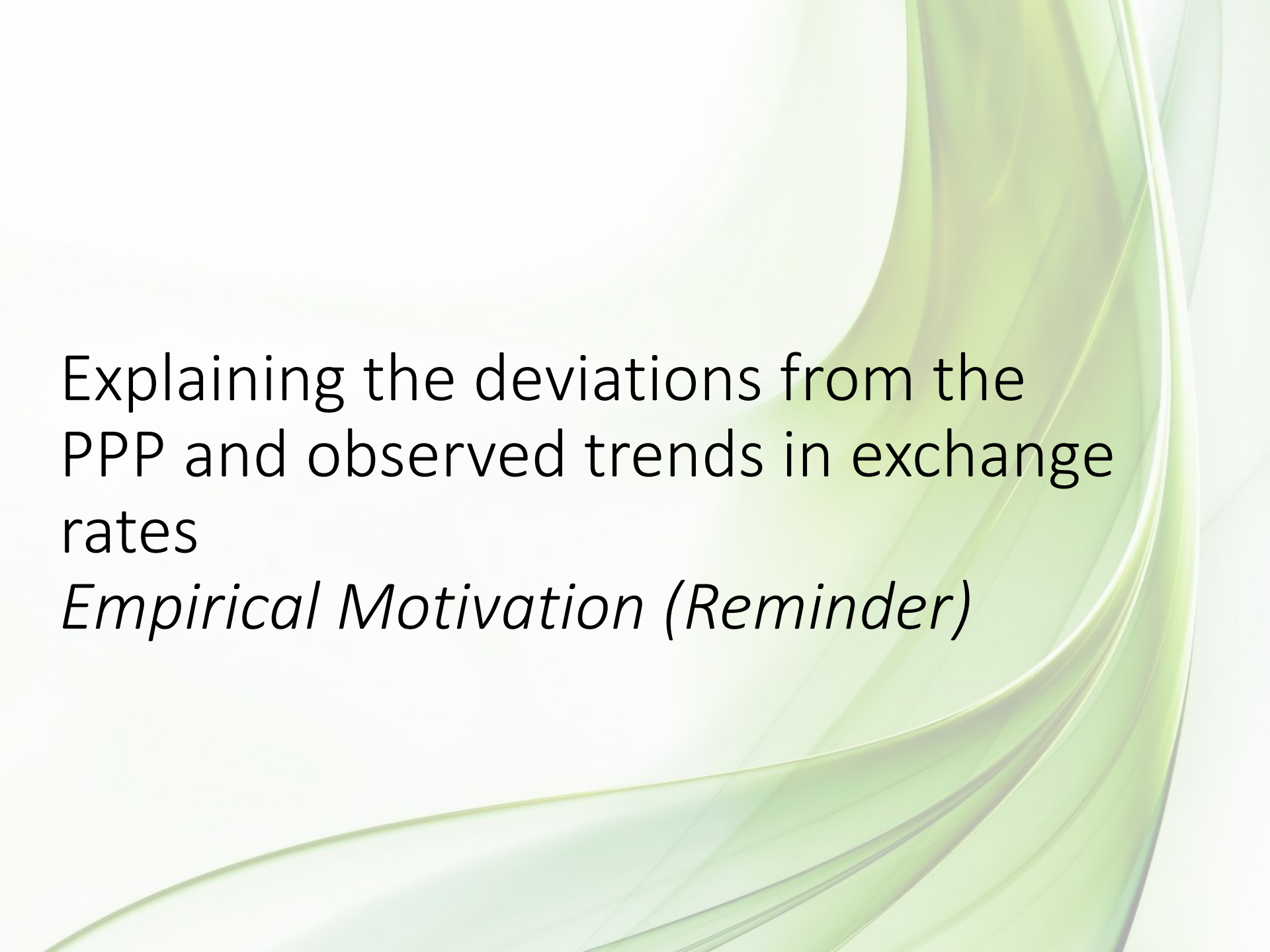


Source: CNB, Arad

Example: Indices of Real Effective Exchange Rates by CNB (Base 2020)



Source: CNB, Arad

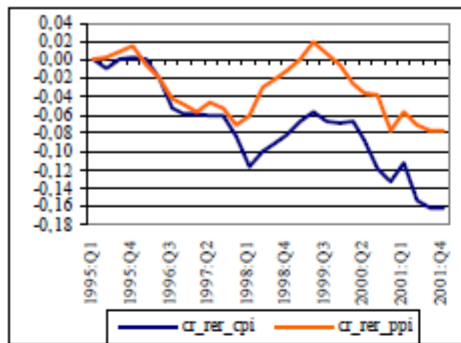
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Explaining the deviations from the
PPP and observed trends in exchange
rates

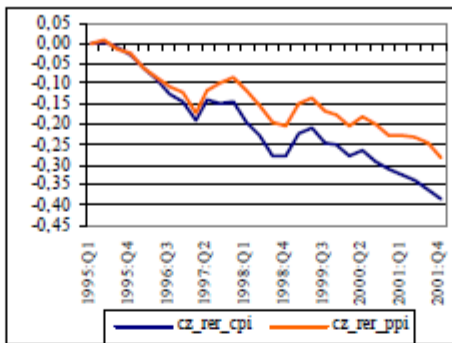
Empirical Motivation (Reminder)

Real Exchange Rates vis-à-vis Germany

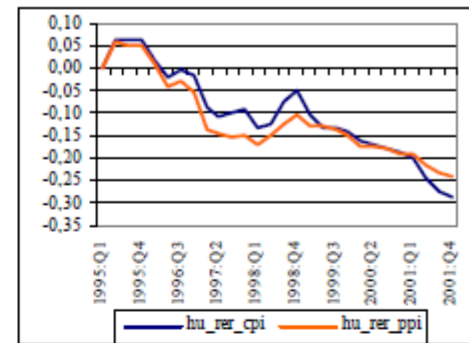
Croatia



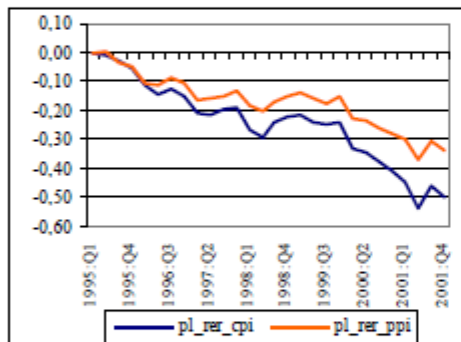
Czech Republic



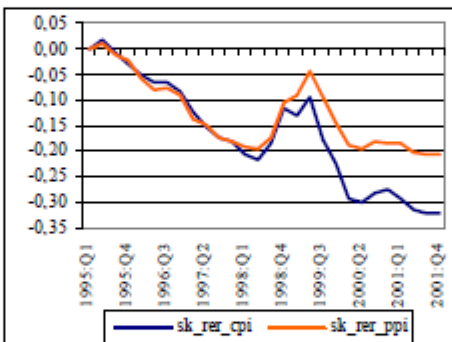
Hungary



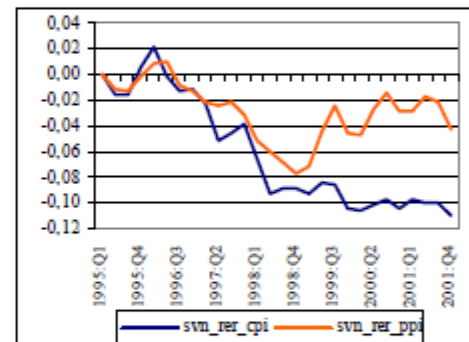
Poland



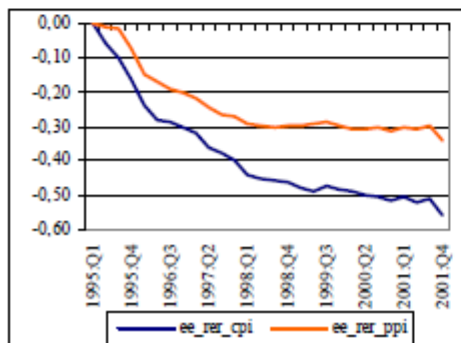
Slovakia



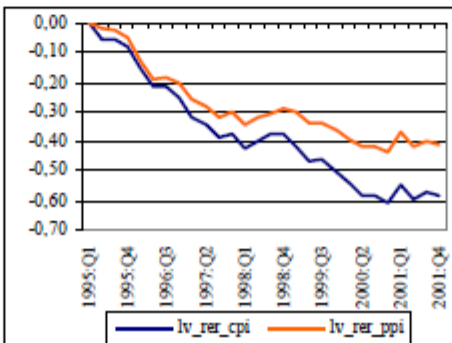
Slovenia



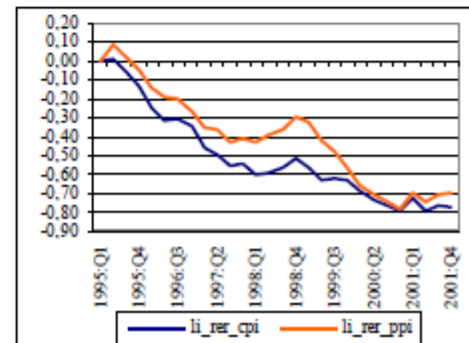
Estonia



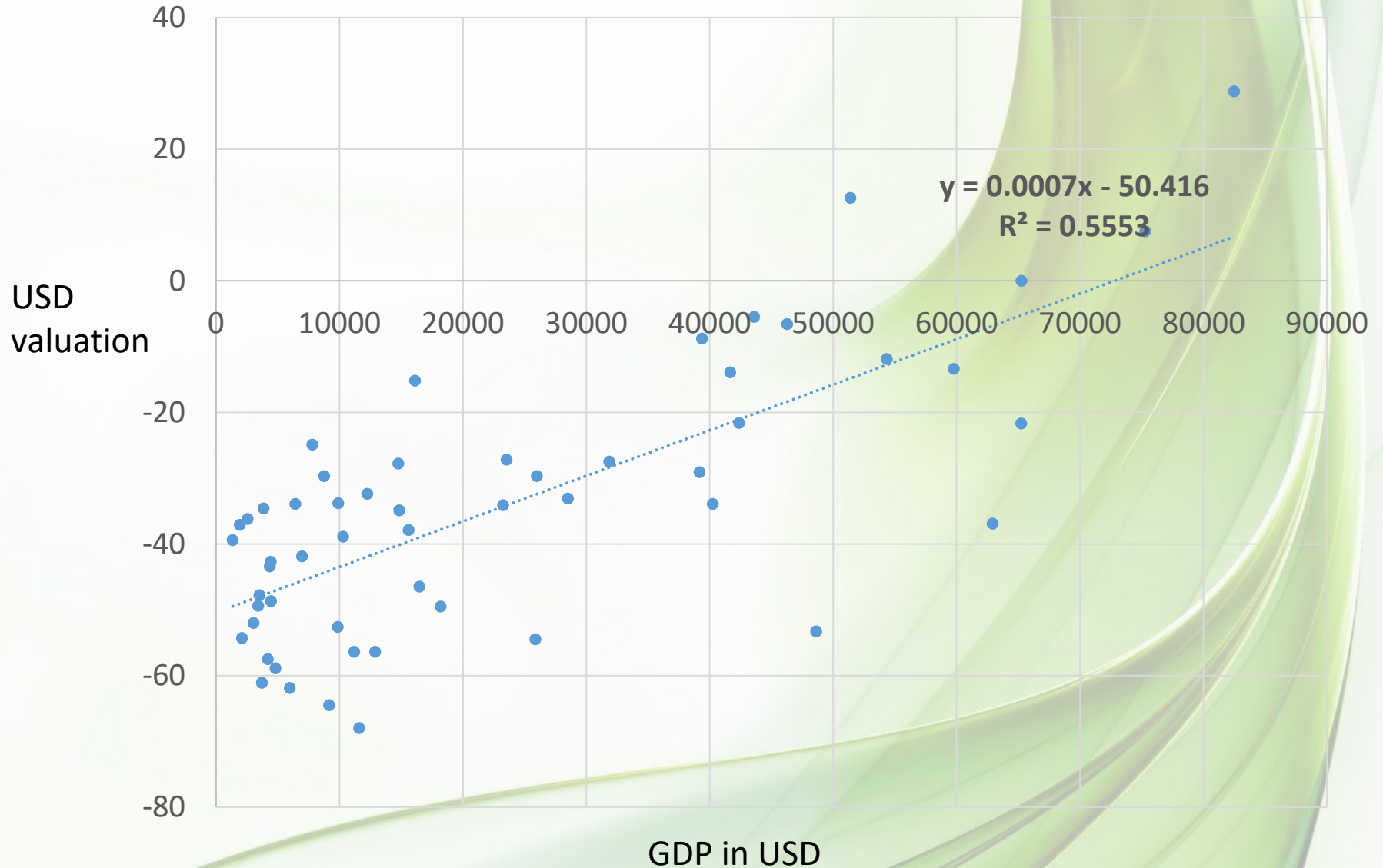
Latvia



Lithuania



Big Mac Indices and the GDP?



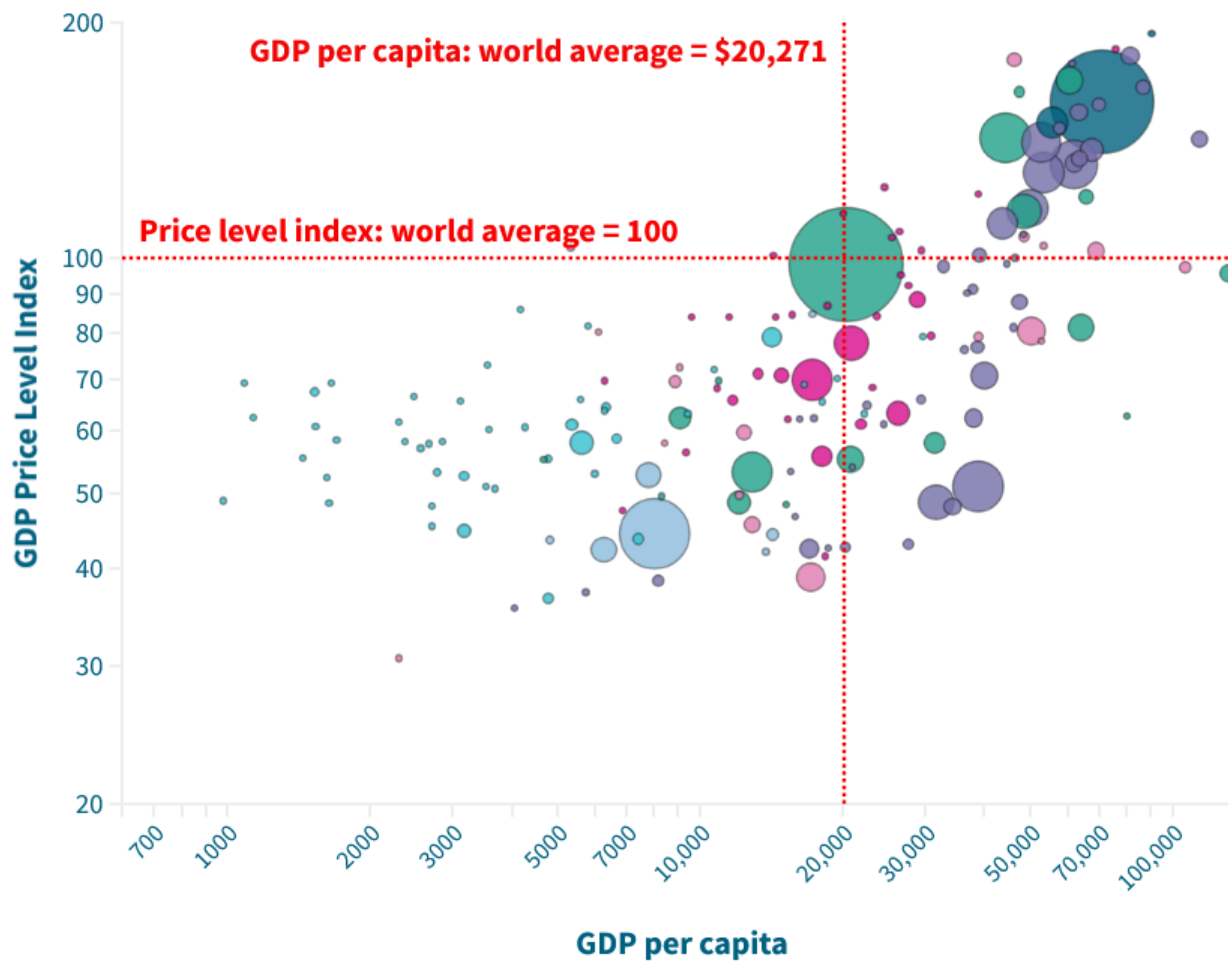
Source of data: Big Mac Index - github

GDP price level index and PPP-based GDP per capita index by economy, 2021

Size of bubble indicates relative size of PPP-based GDP for each economy

Click on chart to view data for each economy

Region ● East Asia & Pacific ● Europe & Central Asia ● Latin America & Caribbean ● Middle East & North Africa
● North America ● South Asia ● Sub-Saharan Africa



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Balassa-Samuelson Effect

Harrod–Balassa–Samuelson effect

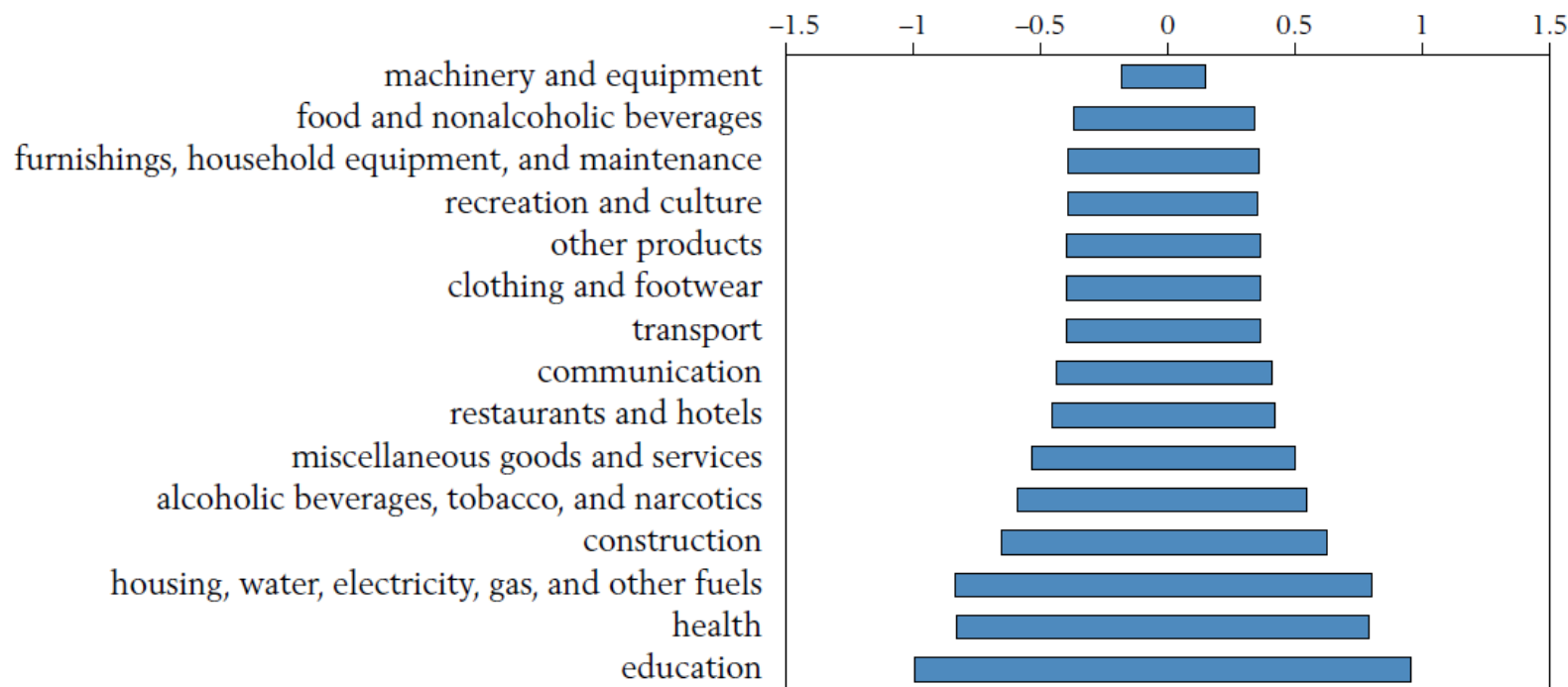
Ricardo–Viner–Harrod–Balassa–Samuelson–Penn–Bhagwati effect

International Price Differences

Example from 2005...

FIGURE 7 CROSS-COUNTRY DIFFERENCES IN PRICES LEVEL INDEXES,
BY PRODUCT GROUPS

(coefficient of variation)



Balassa-Samuelson Effect

- Balassa (1964), Samuelson (1964)
- Explains (some of) the problems of the simple PPP
- Basic assumptions:
 - Two types of goods (and services): Tradables and non-tradables!
 - PPP (absolute) holds only for tradables
 - Price of non-tradables depends on domestic wages
 - Domestic wages are determined in the tradable sector
 - Poorer countries have lower productivity in the tradable sectors (than rich countries), but similar productivity in non-tradable sectors

Balassa – Samuelson Effect

Tradables:

$$\begin{array}{|c|} \hline \text{Foreign price} \\ (P_f^T) \\ \hline \end{array} \times \begin{array}{|c|} \hline \text{Exchange rate} \\ (S) \\ \hline \end{array} = \begin{array}{|c|} \hline \text{Domestic price} \\ (P_d^T) \\ \hline \end{array}$$

Wage:

$$\begin{array}{|c|} \hline \text{Domestic price} \\ (P_d^T) \\ \hline \end{array} \times \begin{array}{|c|} \hline \text{Productivity in tradable sector} \\ (a^T) \\ \hline \end{array} = \begin{array}{|c|} \hline \text{Wages in tradables} \\ (w_d^T) \\ \hline \end{array}$$

Nontradables:

$$\begin{array}{|c|} \hline \text{Wages} \\ (w_d^T) = (w_d^N) \\ \hline \end{array} : \begin{array}{|c|} \hline \text{Productivity in non-tradables} \\ (a^N) \\ \hline \end{array} = \begin{array}{|c|} \hline \text{Domestic price} \\ (P_d^N) \\ \hline \end{array}$$

Our Country: **Lower** Productivity in Tradables (a_d^T)

- Prices of tradables: identical (by assumption)

$$P_d^T = S P_f^T \quad (\text{PPP holds for tradables})$$

- Wages in our country: **lower**

foreign wages: $w_f^T = a_f^T P_f^T$

our wages: $w_d^T = a_d^T P_d^T = a_d^T S P_f^T = S a_d^T P_f^T < S a_f^T P_f^T$
 $w_d^T < w_f^T$

- Prices of non-tradables in our country: **lower**

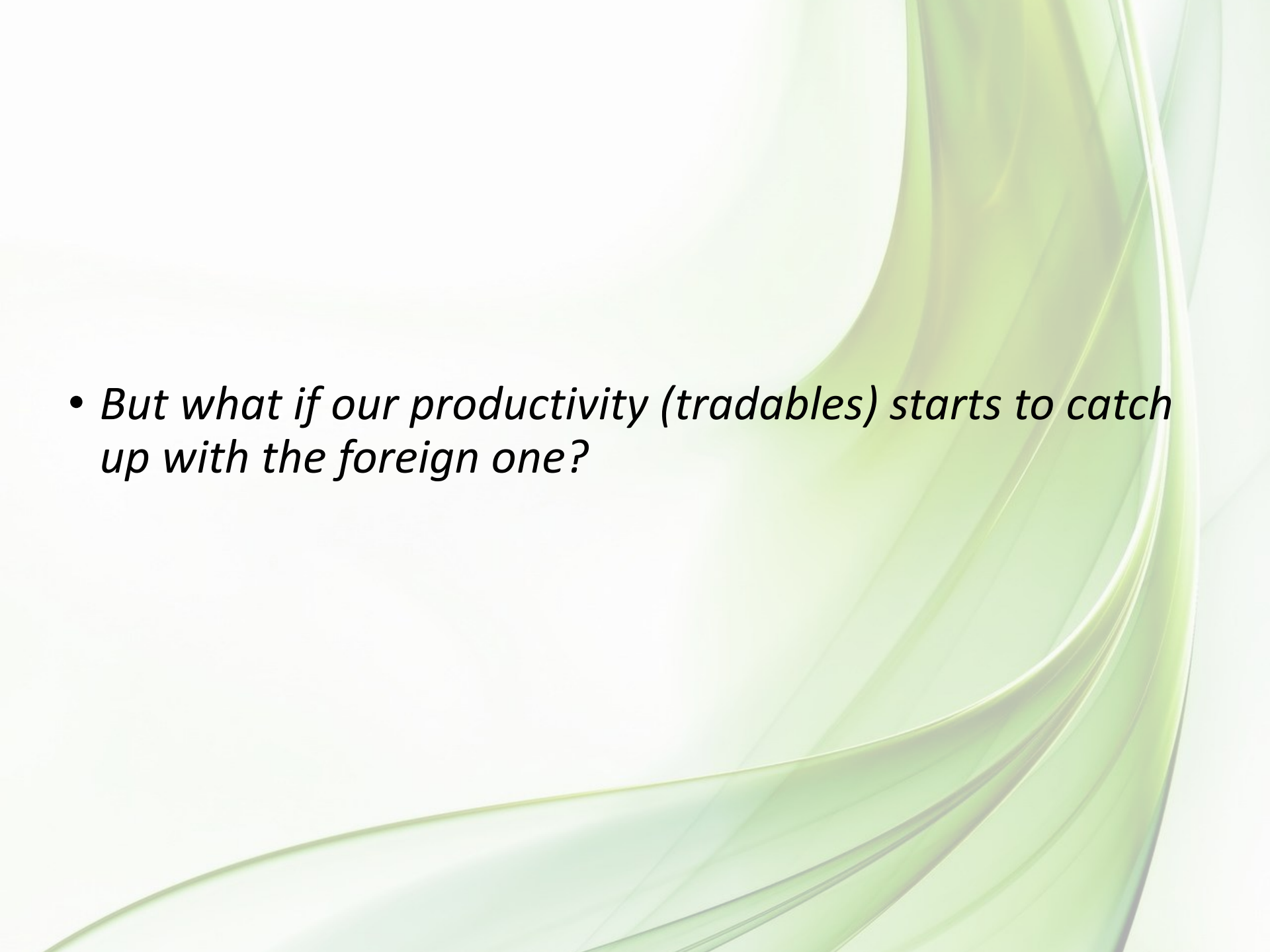
foreign NT prices: $P_f^N = w_f^N / a_f^N = w_f^T / a_f^N = a_f^T P_f^T / a_f^N$

our NT prices: $P_d^N = w_d^N / a_d^N = w_d^T / a_d^N =$
 $P_d^N = S a_d^T P_f^T / a_d^N < S a_f^T P_f^T / a_f^N = P_f^N$

- Average price level in our country must be lower!

Implications:

- Existence of a systematic relationship between productivity in tradables (proxied by GDP per capita) and deviations from the PPP
- Practical issues:
 - Even if our employees in non-tradable sectors (e.g. services) the same productivity as in Germany, their wages will be lower...

- 
- *But what if our productivity (tradables) starts to catch up with the foreign one?*

Dynamic Version:

Our Productivity in Tradables Converges to Foreign
and $S = \text{const.}$

- Prices of our tradables: remain stable

$$P_d^T = S P_f^T \quad (\text{PPP holds for tradables})$$

- Wages in our country: **increase and converge** to foreign wages

$$w_f^T = a_f^T P_f^T \text{ (const.)}$$
$$w_d^T = \uparrow a_d^T P_d^T = \uparrow a_d^T S P_f^T = \uparrow S a_d^T P_f^T \rightarrow S a_f^T P_f^T$$

- Prices of non-tradables: increase and converge to foreign prices of non-tradables

$$P_f^N = w_f^N / a_f^N = w_f^T / a_f^N = a_f^T P_f^T / a_f^N$$
$$P_d^N = w_d^N / a_d^N = w_d^T / a_d^N = \uparrow S a_d^T P_f^T / a_d^N \rightarrow S a_f^T P_f^T / a_f^N$$

- Average price level in our country: increases
 - We are experiencing inflation and real appreciation

Balassa-Samuelson Effect (2)

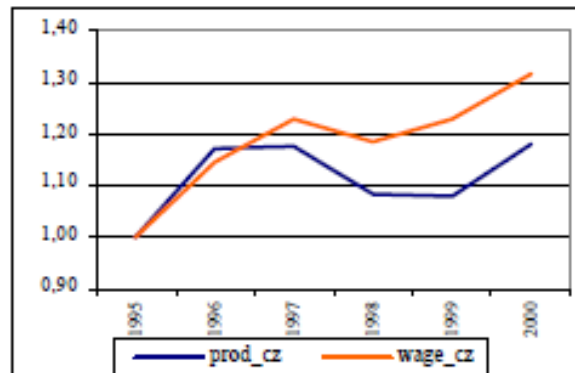
- Implications of the productivity growth in tradable sector
 - Increasing wages in the tradable sector
 - “Wage disease” will lead to increasing wages also in the non-tradable sector
 - If productivity in non-tradable sector does not change, increasing wages will lead to higher prices of non-tradables
- Let us consider two scenarios:
 - I. Constant nominal exchange rate (e.g. pegged exchange rate)
 - Stable prices of tradables, increasing prices of non-tradables
 - There is **inflation** caused by increasing prices of non-tradables
 - II. **Nominal appreciation**
 - Prices of tradable (in domestic currency) decrease, prices of non-tradables are increasing (less than in the case I)
 - Average price level can be constant (zero overall inflation)

Productivity and Real Wages in Industry

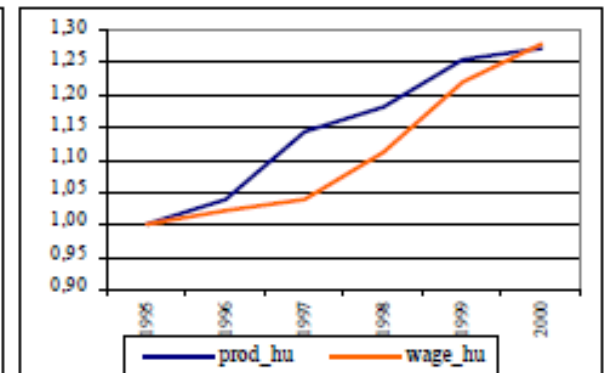
Croatia



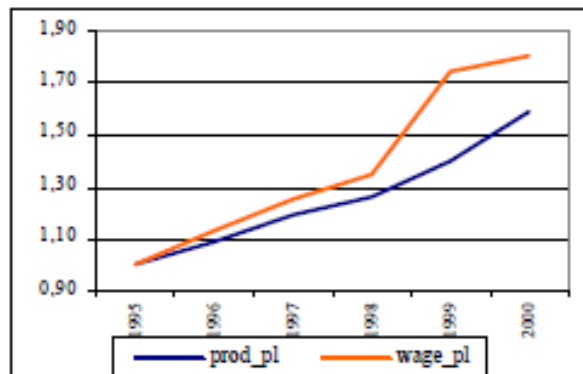
Czech Republic



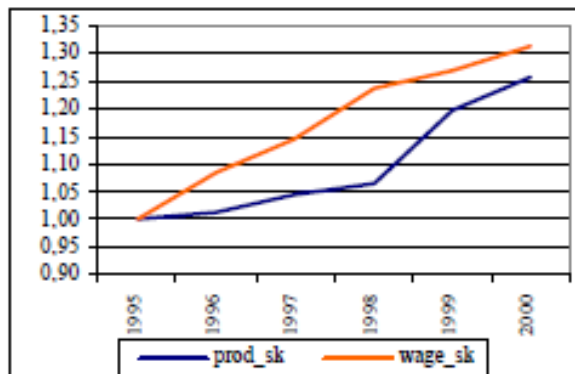
Hungary



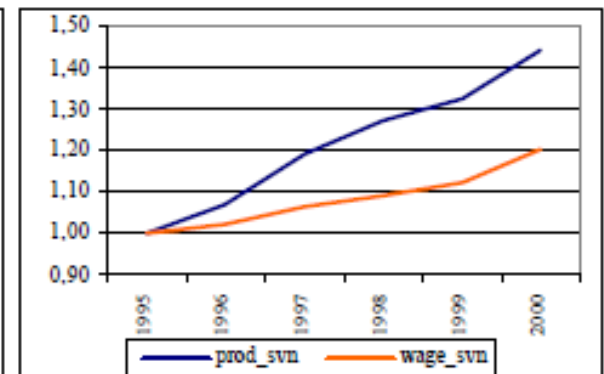
Poland

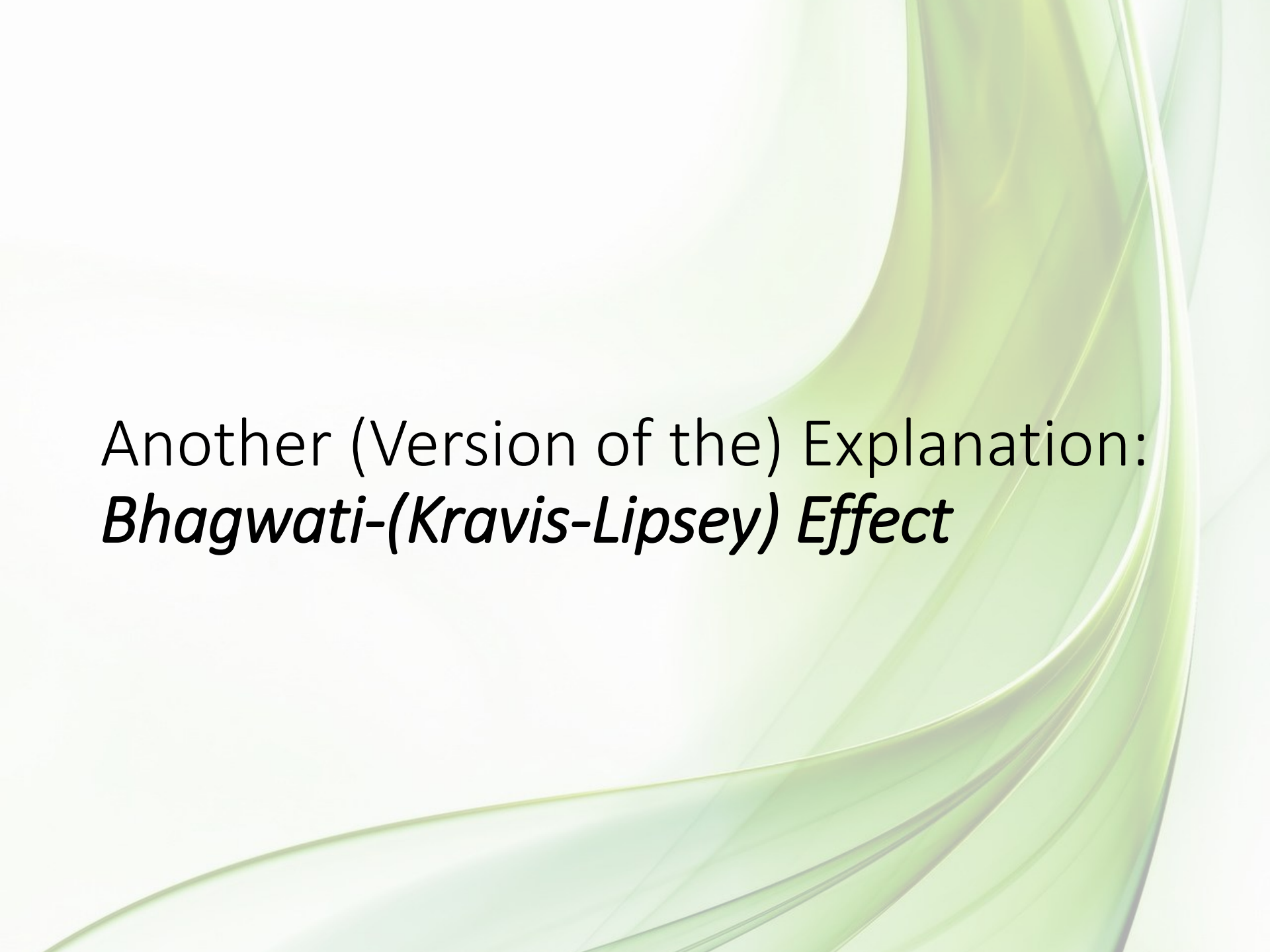


Slovakia



Slovenia





Another (Version of the) Explanation:
Bhagwati-(Kravis-Lipsey) Effect

Explanation via K/L Differences

- Poor countries have **lower capital-labor (K/L) ratios** than rich countries
 - Marginal product of labor in poor countries is therefore lower, therefore wages are lower too
- The production of non-tradable goods (e.g. services) is labor-intensive
- Labor in poor countries is cheaper and because it is intensively used in the production of non-tradable goods → non-tradable goods should be cheaper in poor countries
- Even if PPP holds for tradables, the overall price level should be lower in poor countries
 - Hence again correlation between CPL and GDP per capita

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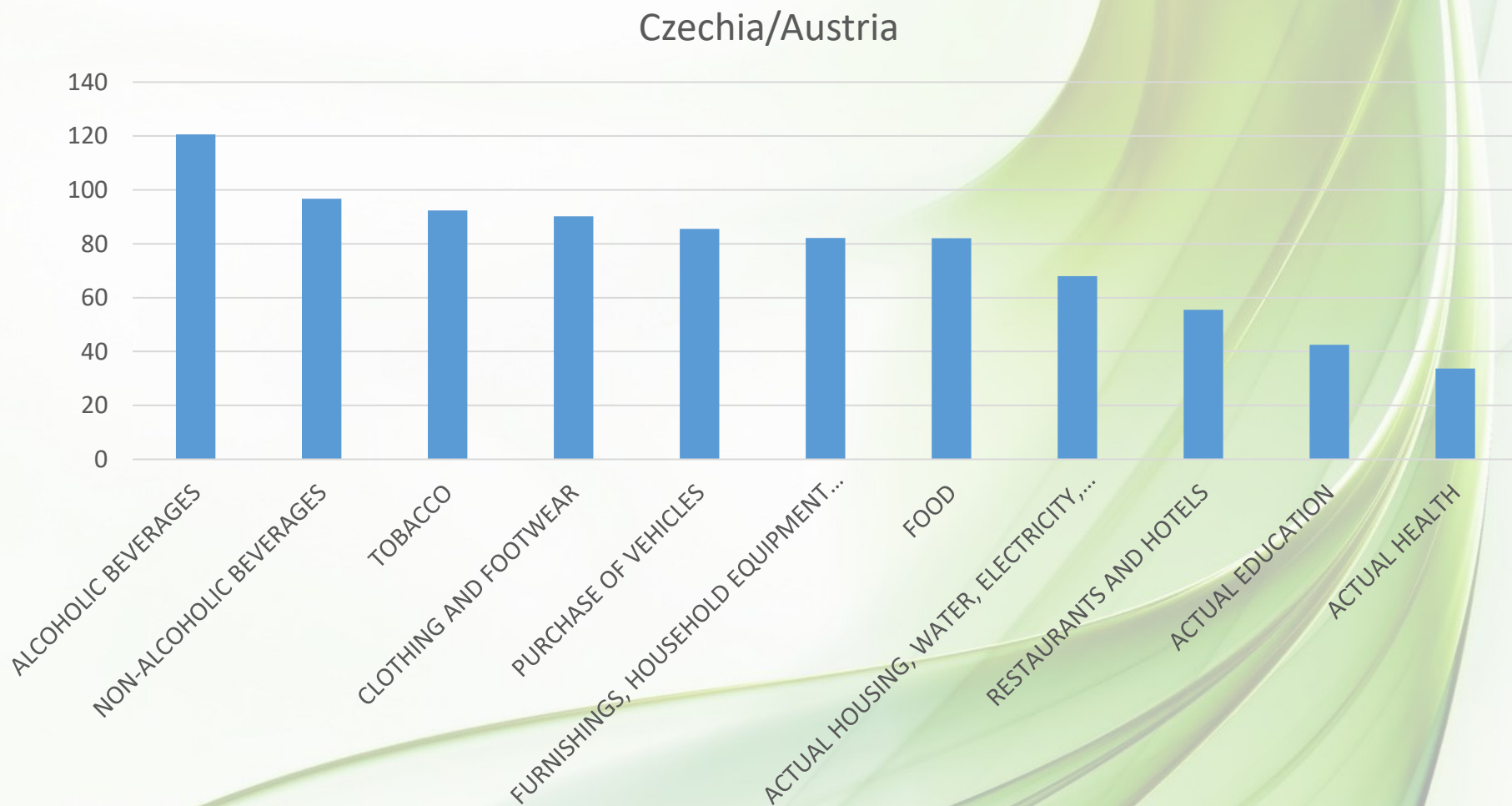
Empirical Relevance of Balassa & Samuelson

BS Effect: Euro and Prices/Inflation in the CEE/Czechia

- Czechia indeed had very different relative prices (e.g. relative to Austria)
- Czechia was indeed experiencing very different speeds of price adjustments for different commodities/services

ICP 2021: Czechia v. Austria

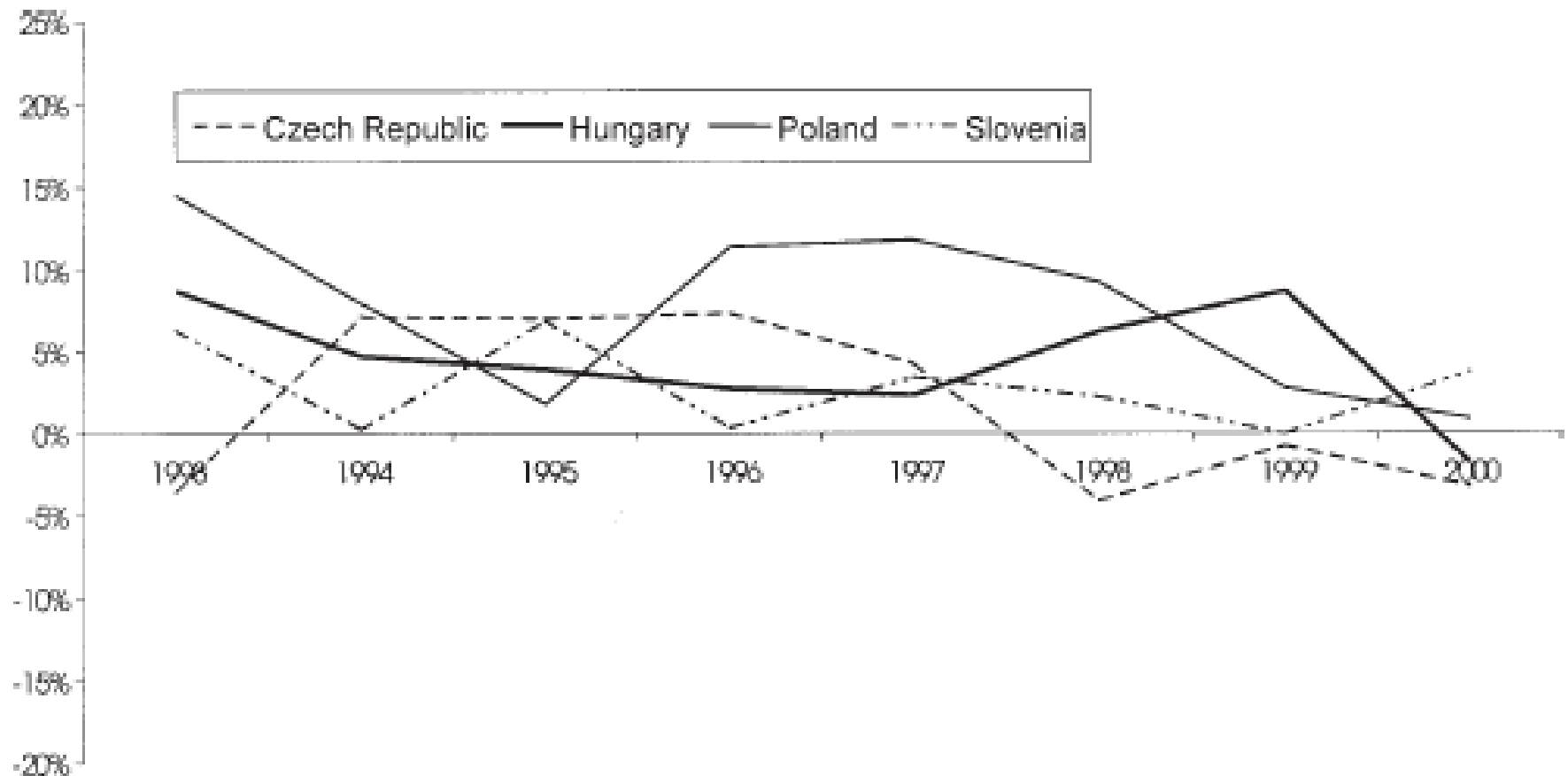
Austrian price level in each category = 100



Source: WB ICP website

Non-tradable prices were really growing faster...

Figure 3. Inflation Differential: Non-Tradables/Tradables, 1993–2000

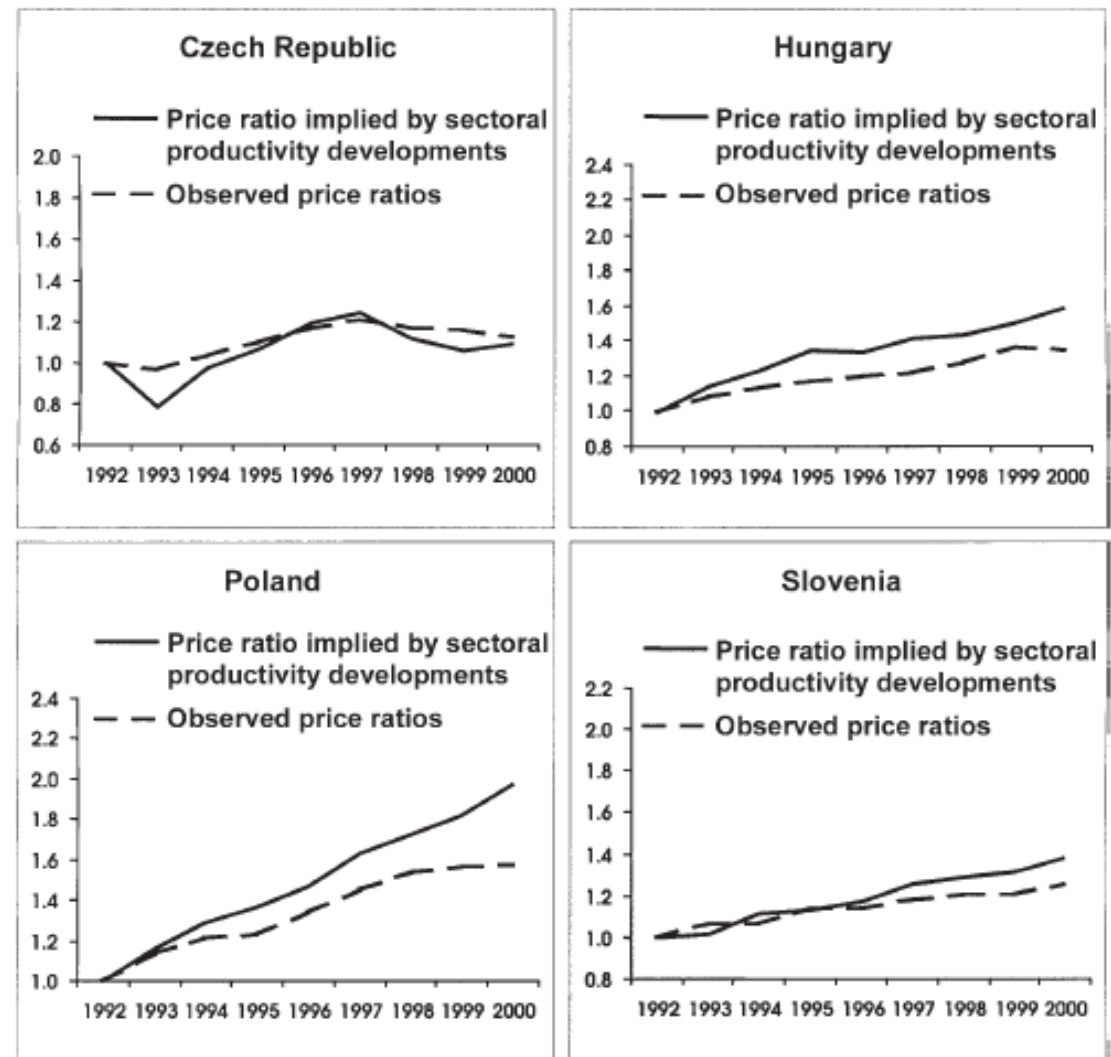


Sources: WIIW and OeNB calculations.

Note: Based on implicit sectoral deflators of gross value added.

But productivities do not predict it perfectly...

Figure 5. Relative Prices: Non-Tradables/Tradables (Price Ratio), 1992 = 1



Sources: WIIW and OeNB calculations.

Notes: Labor productivities were used instead of total productivities (for sectoral productivities). The labor shares γ and δ were computed as the share of total wage sum in gross value added in the respective sectors.

Empirical Tests

- ..the Balassa-Samuelson effect for the transition economies is ‘ ***now well established and powerful*** ’ (Begg et. al ., 2001: ix),
- Egert (2002): analysis of the role of the BS effect in the CR, Hungary, Poland, Slovenia, and Slovakia
 - In the case of CR and SR almost zero estimated impact on inflation differentials in relation to Germany, for Hungary and Poland at most 3.5 perc. points.
 - Other estimates:
 - Sinn & Reutter (2001): 2.88% for the CR, 6.86% for Hungary

Newer Empirical Tests

- Chen, Choi & Devereux (2015):
 - *We test Balassa-Samuelson by quantifying the contribution of non-tradables to cross-sectional price level variation for countries in the International Comparison Program (ICP).*
 - *Our results suggest that **non-tradables explain up to two thirds of price level variation.***
- Bordo et al (2017):
 - Historical data **for over hundred years and 14 countries** is used to estimate the long-run effect of productivity on the real exchange rate.
 - ... the traditional Balassa-Samuelson model is not consistent with these results
 - ... we argue that changes in trade costs over time may affect the impact of productivity on the real exchange rate over time

Some Other Contributions to the Discussion

- Verma, Radheshyam and Nath, Siddhartha and Bhowmick, Chaitali and Yadav, Swastik (2025): *Balassa-Samuelson Effect in Emerging Market Economies- An Empirical Examination*.
 - From their abstract:
 - This study revisits the Balassa-Samuelson (B-S) hypothesis for 16 inflation-targeting emerging market and developing economies (EMDEs) to test whether their inflation differential with the advanced economies (AEs) could be explained through the productivity channel.
 - **The study finds positive and significant impact of total factor productivity (TFP) and labour productivity (LP) growth differentials on inflation differentials between AEs and inflation targeting EMDEs.** The B-S effect is estimated in the range 1.6-2.5 percentage points for India.

Balassa-Samuelson and Policy-Making



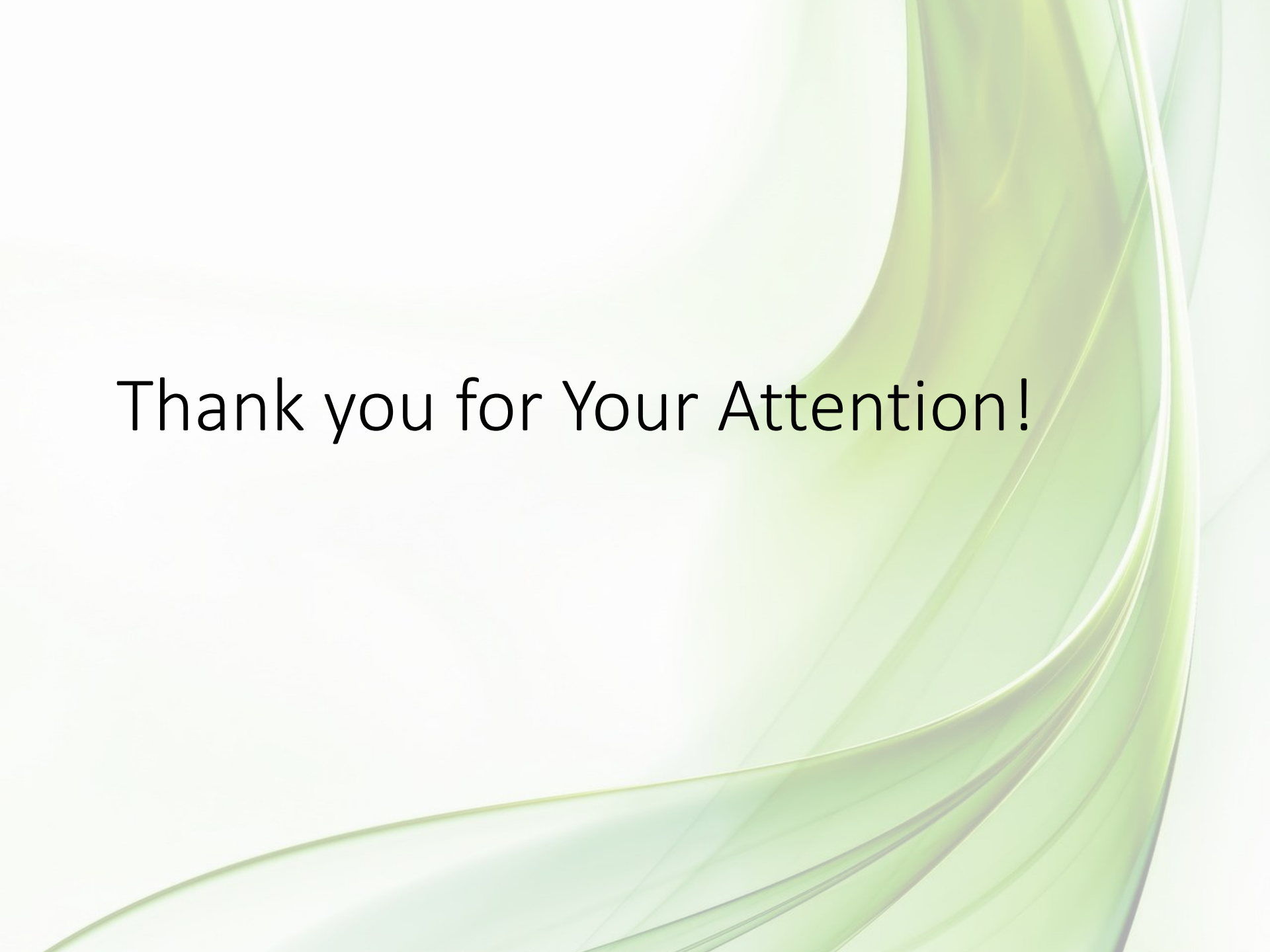
BS Effect: ERM2, Euro and Inflation

- A tight peg to the euro might produce an inflation rate above the Maastricht ceiling
 - This happened in pre-entry Estonia or post-entry Ireland
- Reasons to expect higher inflation rate also after the introduction of Euro
 - But:
 - This type of inflation (in prices of non-tradables) does not have negative effect on competitiveness (it is caused by increasing productivity and productivity should be increasing faster than price level)
 - Inflation caused by this effect should not be too strong (upper limit given by the productivity differential)

Convergence Criteria for Euro Adoption

What is measured:	Price stability	Sound and sustainable public finances	Durability of convergence	Exchange rate stability
How it is measured:	Harmonised consumer price inflation	Government deficit and debt	Long-term interest rate	Exchange rate developments in ERM II
Convergence criteria:	A price performance that is sustainable and average inflation not more than 1.5 percentage points above the rate of the three best performing Member States	Not under excessive deficit procedure at the time of examination	Not more than 2 percentage points above the rate of the three best performing Member States in terms of price stability	Participation in ERM II for at least 2 years without severe tensions, in particular without devaluing against the euro

Source: https://economy-finance.ec.europa.eu/euro/enlargement-euro-area/convergence-criteria-joining_en

The background of the slide features abstract, flowing, organic shapes in various shades of green and white. These shapes curve and sweep across the frame, creating a sense of movement and depth. The colors transition from light, airy whites to vibrant, saturated greens, with some areas appearing more translucent than others.

Thank you for Your Attention!

References

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